Madhya Pradesh (Bhopal lake & fringe)

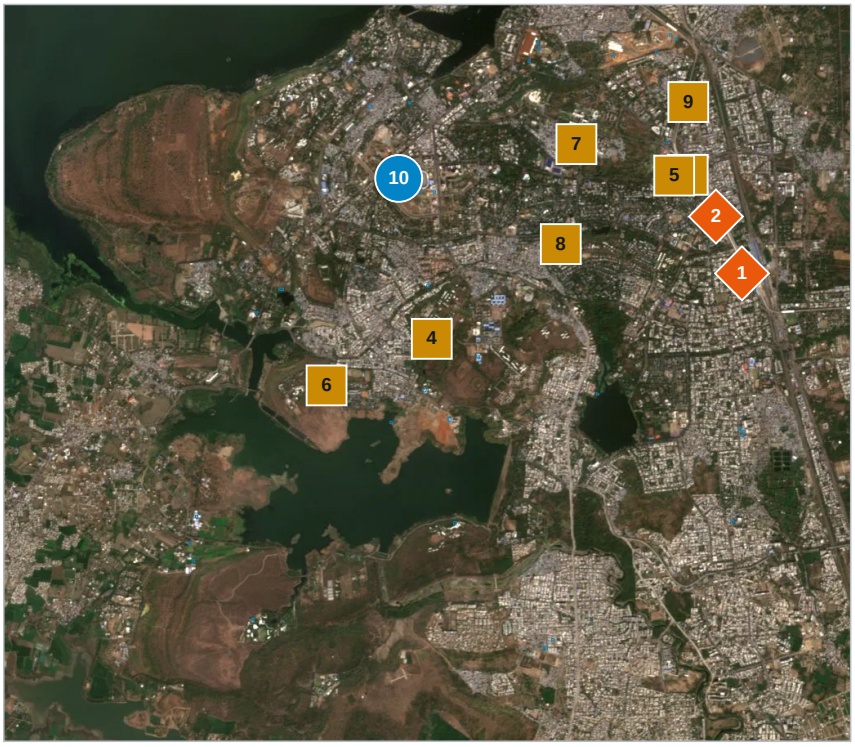
Bhopal (Bhojtal catchment & Kolar corridor), Madhya Pradesh | Before - Sentinel-2 - 2019-01-22 to After - Sentinel-2 - 2025-01-25

What we found

We found 47 real changes covering 8.09 hectares. The biggest and strongest (rank 1) is new construction: 0.73 ha at 23.2213, 77.4382.

By priority: 0 Critical, 2 High, 7 Medium, 38 Low.

The detector first flagged 3,649 candidate areas. The rest were removed as false alarms: change types this analysis does not track, seasonal crop and vegetation cycles, tiny specks and noise, cloud and cloud shadow.



Numbers = rank of the 10 highest-priority changes. Outline colour = priority.

Changes by priority

Critical	0
High	2
Medium	7
Low	38

Totals

Area changed: 8.09 ha

Cropland affected: 0.12 ha (upper bound)

Built-up land affected: 6.31 ha

How to read this report

Rank	Every change has a number. Rank 1 is the highest priority: the largest and strongest change. Priority = size x strength of the change.
Priority colour and shape	Critical (octagon): score 20,000 or more (about 2 ha of strong change); High (diamond): score 5,000 or more; Medium (square): score 1,500 or more; Low (circle): score below 1,500 (small or weak).
Confidence	How sure the system is that the change is real, from 0 to 100 %. Changes marked 'needs a human check' sit between 50 % and 70 %.
What changed	New construction, vegetation cleared, vegetation growth, new water / flooding, water receded, or unexplained change. These come from rules on the satellite colours, not from knowing what is actually on the ground.
Raw detections	Everything the detector first flagged. Most are not real (clouds, crop seasons, shadows, specks), so only changes that pass the filters are listed here.

The 10 highest-priority changes

Start here. Each card says where the change is, how big it is, why it was flagged and what to do.

1

New construction

Rank 1 of 47 | High priority

Size	0.73 ha (7,300 m2)
Confidence	97 %
Location	23.22130, 77.43820 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	built-up index fell (-0.09), vegetation index fell (-0.08), water index rose (+0.11), SWIR/blue ratio (paved surfaces are flatter) fell (-0.76)
What to do	Check soon. A substantial change that is probably real.

Before



After



2

New construction

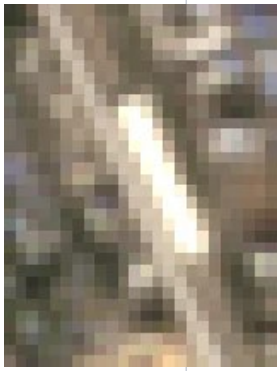
Rank 2 of 47 | High priority

Size	0.75 ha (7,500 m2)
Confidence	97 %
Location	23.22770, 77.43503 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	built-up index fell (-0.07), vegetation index fell (-0.08), water index rose (+0.11), SWIR/blue ratio (paved surfaces are flatter) fell (-0.60)
What to do	Check soon. A substantial change that is probably real.

Before



After



3

New construction

Rank 3 of 47 | Medium priority

Size	0.55 ha (5,450 m2)
Confidence	97 %
Location	23.23240, 77.43161 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	built-up index fell (-0.08), vegetation index fell (-0.10), water index rose (+0.12), SWIR/blue ratio (paved surfaces are flatter) fell (-0.76)
What to do	Check when you can, for example on a routine inspection.

Before



After



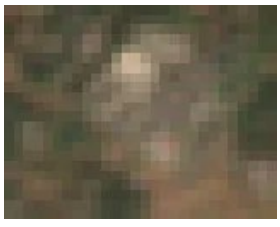
4

New construction

Rank 4 of 47 | Medium priority

Size	0.43 ha (4,350 m2)
Confidence	97 %
Location	23.21392, 77.40054 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	built-up index fell (-0.25), vegetation index fell (-0.07), water index rose (+0.10), SWIR/blue ratio (paved surfaces are flatter) fell (-1.48)
What to do	Check when you can, for example on a routine inspection.

Before



After



5

New construction

Rank 5 of 47 | Medium priority

Size	0.42 ha (4,250 m2)
Confidence	97 %
Location	23.23232, 77.42997 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	built-up index fell (-0.12), SWIR/blue ratio (paved surfaces are flatter) fell (-0.58)
What to do	Check when you can, for example on a routine inspection.

Before



After



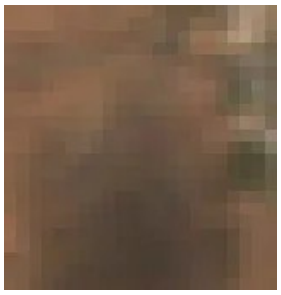
6

New construction

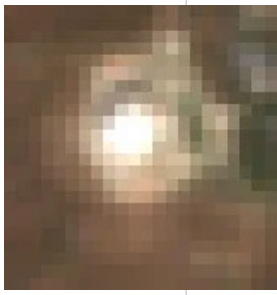
Rank 6 of 47 | Medium priority

Size	0.30 ha (3,000 m2)
Confidence	98 %
Location	23.20867, 77.38777 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	built-up index fell (-0.27), water index rose (+0.13), SWIR/blue ratio (paved surfaces are flatter) fell (-2.46)
What to do	Check when you can, for example on a routine inspection.

Before



After



7

New construction

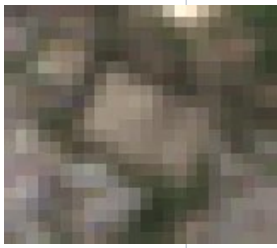
Rank 7 of 47 | Medium priority

Size	0.30 ha (3,000 m2)
Confidence	95 %
Location	23.23587, 77.41810 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	built-up index rose (+0.15), vegetation index rose (+0.08), water index fell (-0.08), SWIR/blue ratio (paved surfaces are flatter) rose (+0.96)
What to do	Check when you can, for example on a routine inspection.

Before



After



8

New construction

Rank 8 of 47 | Medium priority

Size	0.20 ha (2,000 m2)
Confidence	98 %
Location	23.22460, 77.41620 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	built-up index fell (-0.15), SWIR/blue ratio (paved surfaces are flatter) fell (-0.72)
What to do	Check when you can, for example on a routine inspection.

Before



After



9

New construction

Rank 9 of 47 | Medium priority

Size	0.18 ha (1,800 m2)
Confidence	98 %
Location	23.24061, 77.43168 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	built-up index fell (-0.15), vegetation index fell (-0.14), water index rose (+0.18), SWIR/blue ratio (paved surfaces are flatter) fell (-1.29)
What to do	Check when you can, for example on a routine inspection.

Before



After



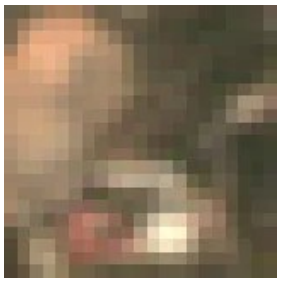
10

New construction

Rank 10 of 47 | Low priority

Size	0.17 ha (1,750 m2)
Confidence	97 %
Location	23.23200, 77.39653 (open in Google Maps)
What it is	Land that was open (vegetation or bare soil) now looks built-up: roofs, roads or concrete.
Why flagged	built-up index fell (-0.18), vegetation index fell (-0.09), water index rose (+0.13), SWIR/blue ratio (paved surfaces are flatter) fell (-1.67)
What to do	Small or weak change. Often noise: confirm before acting.

Before



After



All 47 changes found

Every change that survived the false-alarm filters, in priority order (rank 1 is the highest). A * after the confidence means it needs a human check. The CSV and GeoPackage downloads hold the same list with every attribute.

Rank	Priority	What changed	Size (ha)	Confidence	Latitude	Longitude
1	High	New construction	0.73	97 %	23.22130	77.43820
2	High	New construction	0.75	97 %	23.22770	77.43503
3	Medium	New construction	0.55	97 %	23.23240	77.43161
4	Medium	New construction	0.43	97 %	23.21392	77.40054
5	Medium	New construction	0.42	97 %	23.23232	77.42997
6	Medium	New construction	0.30	98 %	23.20867	77.38777
7	Medium	New construction	0.30	95 %	23.23587	77.41810
8	Medium	New construction	0.20	98 %	23.22460	77.41620
9	Medium	New construction	0.18	98 %	23.24061	77.43168
10	Low	New construction	0.17	97 %	23.23200	77.39653
11	Low	New construction	0.15	98 %	23.24662	77.41362
12	Low	New construction	0.14	98 %	23.21994	77.43954
13	Low	New construction	0.23	95 %	23.17920	77.41432
14	Low	New construction	0.14	98 %	23.19321	77.40336
15	Low	Water receded	0.18	96 %	23.21948	77.38226
16	Low	New construction	0.21	95 %	23.22263	77.43853
17	Low	New construction	0.14	97 %	23.20488	77.40272
18	Low	New construction	0.12	99 %	23.23989	77.43142
19	Low	New construction	0.14	97 %	23.21944	77.43972
20	Low	New construction	0.12	97 %	23.20815	77.39974
21	Low	New construction	0.12	97 %	23.21194	77.40615
22	Low	New construction	0.12	97 %	23.24003	77.39298
23	Low	New construction	0.09	99 %	23.24576	77.42254
24	Low	New construction	0.09	99 %	23.18233	77.37892
25	Low	New construction	0.09	98 %	23.24735	77.41339
26	Low	New construction	0.09	98 %	23.22448	77.41551
27	Low	New construction	0.09	98 %	23.24045	77.39792
28	Low	New construction	0.09	98 %	23.24520	77.41237
29	Low	New construction	0.09	98 %	23.24809	77.42952
30	Low	New construction	0.09	98 %	23.24763	77.43019
31	Low	New construction	0.09	98 %	23.19336	77.43702
32	Low	New construction	0.09	98 %	23.21984	77.40009
33	Low	New construction	0.09	97 %	23.23041	77.39911
34	Low	New construction	0.09	97 %	23.23609	77.42911
35	Low	New construction	0.09	97 %	23.20456	77.39562
36	Low	Water receded	0.09	97 %	23.19093	77.37126
37	Low	Water receded	0.09	97 %	23.20770	77.42068
38	Low	New construction	0.09	97 %	23.18006	77.41520
39	Low	New construction	0.09	96 %	23.20382	77.43790
40	Low	New construction	0.09	96 %	23.20300	77.43818
41	Low	New construction	0.09	96 %	23.21681	77.37933
42	Low	New construction	0.09	96 %	23.24622	77.42782
43	Low	New construction	0.09	96 %	23.18894	77.37161
44	Low	New construction	0.09	96 %	23.19390	77.37209
45	Low	New construction	0.09	96 %	23.20354	77.43838
46	Low	New construction	0.09	95 %	23.22312	77.43864
47	Low	New construction	0.09	95 %	23.23048	77.40087

Data sources

- Copernicus Sentinel-2 L2A (ESA), via the Element84 Earth Search COG bucket
- ESA WorldCover 2021 v200 (10 m), CC-BY 4.0

Limits - what this report cannot tell you

- Optical imagery cannot see through cloud; flagged areas with cloud are removed, not analysed.
- Change type comes from spectral index rules (no training); check critical changes on the ground.
- 10 m pixels cannot resolve single small buildings or people.
- Rivers, sandbars and lake edges move between dates and can be flagged as change; bare soil in dry regions can look like construction. Check flagged areas near water.
- Land cover is ESA WorldCover 2021, newer than the before image (2019): land already cleared or under construction can be mislabelled as cropland, so treat cropland as an upper bound
- Buildings and road segments need OpenStreetMap, which was not reachable when this result was built: shown as unknown, not zero
- This report points to where to look. Confirm important changes on the ground before acting.